Myopia in phonologically conditioned affix order in Yidiɲ

Phonologically conditioned affix order in Yidiɲ Yidiɲ has a total of four derivational affixes: a transitivizing suffix (COMIT/CAUS), an intransitivizing suffix (INTRANS) and two aspectual affixes marking GOING and COMING aspect. The relative order of these affixes is partially influenced by the syllable structure of the root. The GOING suffix precedes the CAUS/COMIT suffix with bisyllabic roots in (1-a). With a trisyllabic root, however, the order of the two affixes is reversed, as in (1-b). Dixon (1977) connects the variable order to the GOING suffix, which has the property of always lengthening the preceding vowel. According to Dixon (1977), the affixes in (1) are ordered in such a way that the GOING suffix lengthens an even-numbered syllable. This empirical generalization builds on strong evidence that Yidiɲ builds binary feet from the left (Dixon 1977; Bowern et al. 2013; Round 2017). Specifically, the GOING syllable lengthens the second syllable in (1-a) and the fourth syllable in (1-b).

(1) Placement of GOING and CAUS/COMIT (Dixon 1977, 228)

a. magi-ːri-ŋa-l-ŋ
[(ma.giː)(ɻi.ŋal)]
climb.up-GOING-COMIT/CAUS-CL-PRES

b. maɖinda-ŋa-ːli-ŋ
[(ma.ɖin)(da.ŋaː)liŋ]
walk.up-COMIT/CAUS-GOING-PRES

c.*[(ma.ɖin)(daː.li)ŋaŋ]

The INTRANS suffix is phonologically similar to the GOING suffix as it also lengthens the preceding syllable, as shown in (2-a). In (2-b), it combines with a trisyllabic root and a monosyllabic, inflectional PURP suffix. In short, the context is phonologically similar to (1-b), where the placement of the monosyllabic affix after the root allows the GOING suffix to lengthen an even-numbered syllable. In (2-b), however, affix reordering does not take place (see (2-c)). Rather, the INTRANS suffix does not lengthen its preceding syllable. Instead, the suffix is lengthened by an independent phonological process which lengthens penultimate syllables of words with an odd number of syllables.

(2) Relative order of INTRANS and PURP (Dixon 1977, 218)

a. ɖara-ːɖi-ŋada-ŋ
[(ɖa.raː).(ɖi.ŋa)daŋ]
stand-INTRANS-COMING-PRES

b. wuŋaba-ːɖi-na
[(wu.ŋa).(ba.ɖiː).na]
hunt.for-INTRANS-PURP

c.*[(wu.ŋa).(ba.naː).ɖi]

d.*[(wu.ŋa).(baː.ɖi).na]

Affix order in Yidiɲ is myopic I claim that the alternation in the relative order in of suffixes in (1) is an instance of phonologically conditioned affix order. Moreover, I suggest that inflectional affixes, such as PURP *na* are irrelevant for reordering because they enter the word-building process at a later point in the derivation, thus counter-feeding phonologically conditioned affix placement. Hence, the phenomenon MYOPIC: the inflectional affixes have to right phonological shape to optimize the morphological structure, but enter the word too late to participate. I analyze this puzzle in Stratal Optimality Theory. I assume that the derivational affixes (INTRANS, CAUS/COMIT, GOING, COMIT) are concatenated at stem-level. Morphemes present at this level will be syllabified and phonologically conditioned affix order applies to optimize the syllable structure. Inflectional affixes, however, attach at word-level. At this level, syllabification applies, as well as penultimate lengthening and affix shortening. The following table illustrates the cyclic make-up of the word.

Slot 1	Slot 2	Slot 3	Slot 4	Slot 5	Slot 6
root	derivational affixes			class marker	inflection
	CAUS/COMIT *-ŋa/-maŋa/-maŋa -l*			*-n/-l/-ɻ*	PST *-ɲu*
	INTRANS *-ːɖi -n*				NONPST *-ŋ/-l/-ɻ*
	GOING *-ŋali/-ːli/-ːɻi -n*				IMP −[-APPROX]
	COMING *-ŋada/-ːda/-ːda -n*				PURP *-na*
					DAT.SUBORD *-ɲunda*
					CAUS.SUBORD *-ɲum*
					LEST *-ɖi*

stem-level: phonologically conditioned affix order & syllabification

word-level: affix shortening, penultimate lengthening & syllabification

Evidence for Stratality Evidence for a stratal word structure comes from a phonological process that deletes the final syllables of words with an odd number of syllables. In (3-), the bisyllabic root combines with a monosyllabic suffix. However, the input does not surface as a trisyllabic word as in (3-b), but as a bisyllabic word in (3-a). Concretely, the vowel of the affix is deleted, the onset of the monosyllabic suffix becomes the coda of the preceding syllable and the vowel of the preceding syllable is lengthened.

(3) Affix shortening (Dixon 1977, 44)

a. /gali-ɲu/
[(ga.li:ɲ)]
go-ERG

b.*[(ga.li).ɲu]

c. /gindanu/
[(gin.da:n)]
'moon.ABS'

As shown in (3-c), final shortening also applies to non-suffixed roots. However, shortening is only possible when shortening creates a licit coda. In (4), deleting the vowel would create an illicit coda cluster ((4-b)). Deleting one of the coda consonant is not possible ((4-c)). As a result, final shortening does not apply, the word remains trisyllabic and penultimate lengthening applies. Final shortening of inflectional suffixes behaves differently in this regard. As shown in (5), consonants of the suffix are deleted to create an even-syllabled word with a licit coda. The difference between root consonants and affix consonants provides evidence for cyclicity: I assume that final deletion applies at word-level. At this point of the derivation, roots have already passed a cycle of syllabification, while affix consonants have not. Deletion of consonants is only possible when the consonant has not been parsed into a syllable.

(4) Final root deletion (Round 2017, 64)

a. /binarŋa/
[(bi.na:r)ŋa]
'warn-CL-IMP'

b.*[bina:rŋ]

c.*[bina:r]

(5) Final suffix deletion (Round 2017, 64)

a. /binarŋa-l-ɲunda/
[(bi.nar)(ŋal.ɲu:n)]
'warn-CL-DAT.SUBORD'

b.*[(bi.nar)(ŋal.ɲu:nd)]

c.*[(bi.nar)(ŋalɲu:n).da)]

Discussion In this paper, I offer a unified account of several morphophonological phenomena in Yidiɲ: a stratal analysis neatly explains why inflectional affixes a) are invisible to phonologically conditioned affix order and b) behave differently in final shortening.

References Bowern, C, B. Alpher & E. Round (2013): 'Yidiny stress, length, and truncation reconsidered. • Dixon, R. (1977) 'A grammar of Yidiɲ' • Round, E. (2017) 'Phonological exceptionality is localized to phonological elements: The argument from learnability and Yidiny word-final deletion'